

Introduction to version control

INTRODUCTION TO GIT



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What is version control?

- Processes and systems to manage changes to files, programs, and directories



What should be version controlled?

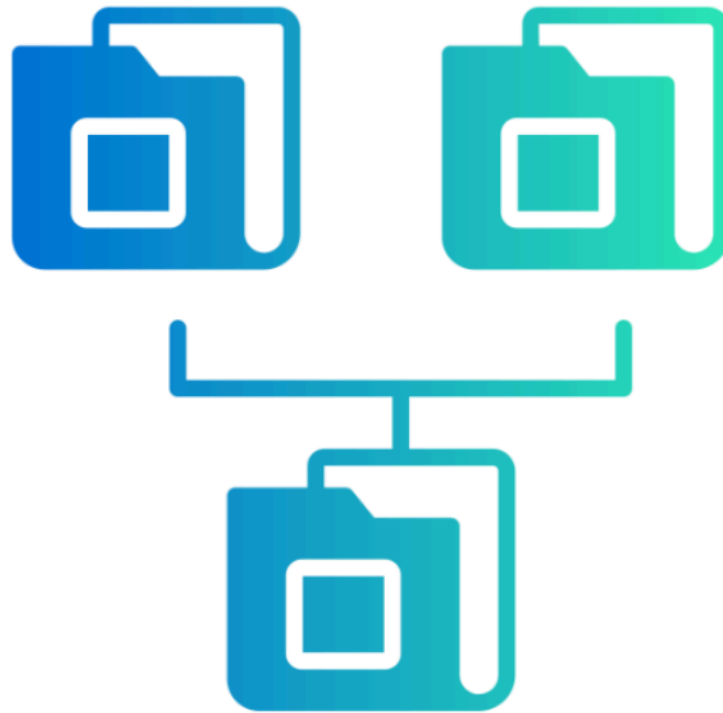
- Version control is useful for anything that:
 - changes over time, or
 - needs to be shared



```
31 def __init__(self, settings):
32     self.file = None
33     self.fingerprints = set()
34     self.logdupes = True
35     self.debug = debug
36     self.logger = logging.getLogger(__name__)
37     if path:
38         self.file = open(os.path.join(path, 'requests.log'),
39                          'a')
40         self.file.seek(0)
41         self.fingerprints.update([fp for fp in self.fingerprints])
42
43 @classmethod
44 def from_settings(cls, settings):
45     debug = settings.getbool('DEBUG', False)
46     return cls(job_dir(settings), debug)
47
48 def request_seen(self, request):
49     fp = self.request_fingerprint(request)
50     if fp in self.fingerprints:
51         return True
52     self.fingerprints.add(fp)
53     if self.file:
54         self.file.write(fp + os.linesep)
55
56 def request_fingerprint(self, request):
57     return request_fingerprint(request)
```

What can version control do?

- Track files in different states
- Combine different versions of files
- Identify a particular version
- Revert changes



Why is version control important?



Why is version control important?



Introducing Git



- Popular version control system for software development and data projects
- Open source
- Scalable

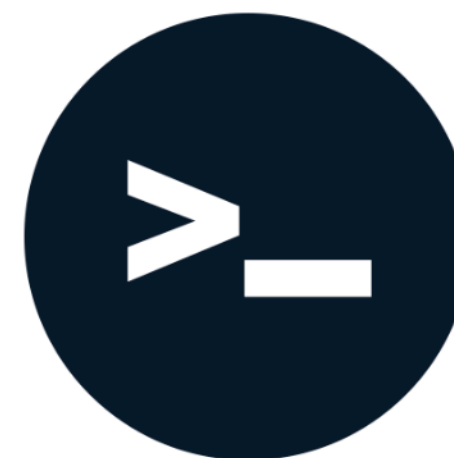
Benefits of Git

- Git stores everything, so nothing is lost
- We can compare files at different times
- See what changes were made, by who, and when
- Revert to previous versions of files!



Using Git

- Git commands are run on the **shell**, also known as the *terminal*
- The shell:
 - is a program for executing commands
 - can be used to easily preview or inspect files and directories
- Directory = folder



Documents



Mental Health in
Tech Project

Useful terminal commands

```
pwd
```

```
/home/rep1/Documents
```

```
ls
```

```
archive    finance.csv  finance_data_clean.csv  finance_data_modified.csv
```

Changing directory

```
cd archive
```

```
pwd
```

```
/home/repl/Documents/archive
```

Checking Git version

```
git --version
```

```
git version 2.46.0
```


Let's practice!

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Creating repos

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funding.doc

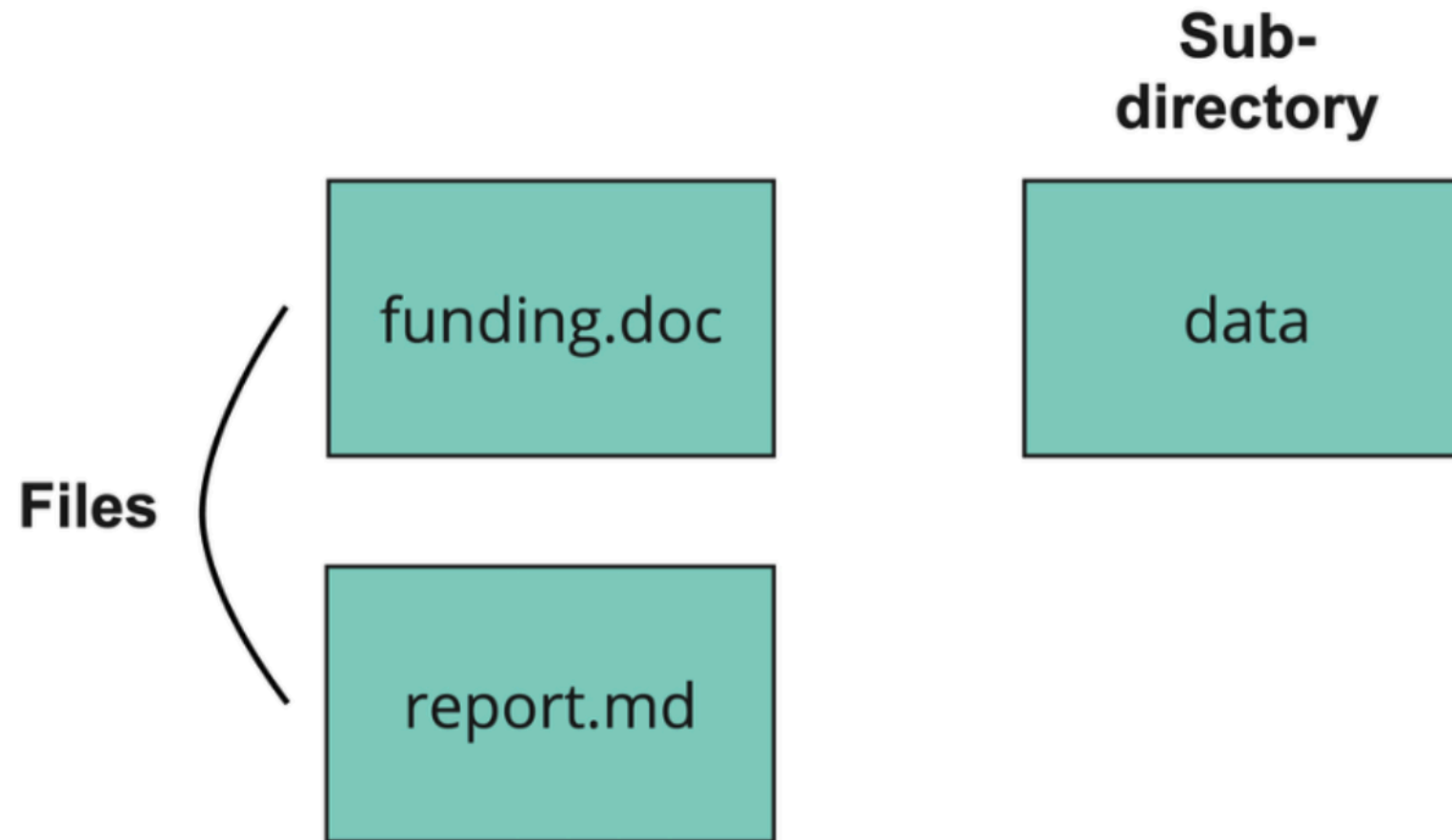
data

.git

report.md

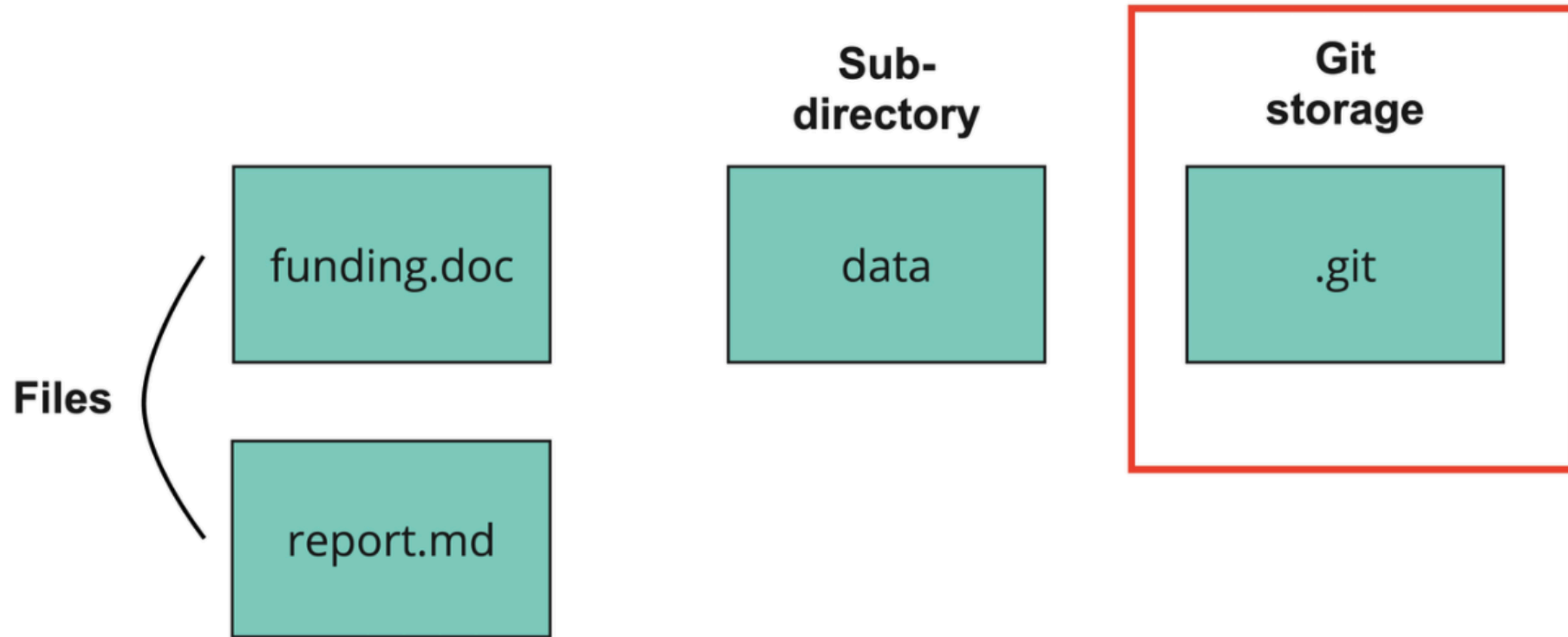
What is a Git repo?

- Git repo = directory containing files and sub-directories



What is a Git repo?

- Git repo = directory containing files and sub-directories, and Git storage



Do not edit `.git` !

Benefits of repos

- Systematically track versions
- Revert to previous versions
- Compare versions at different points in time
- Collaborate with colleagues



Creating a new repo

```
git init mental-health-workspace
```

```
cd mental-health-workspace
```

```
git status
```

```
On branch main
```

```
No commits yet
```

```
nothing to commit (create/copy files and use "git add" to track)
```

Converting a project into a repo

- Convert an existing directory into a Git repo

```
git init
```

```
Initialized empty Git repository in /home/repl/mental-health-workspace/.git/
```

What is being tracked?

```
git status
```

```
On branch main
```

```
No commits yet
```

```
Untracked files:
```

```
  (use "git add <file>..." to include what will be committed)
```

```
    data/
```

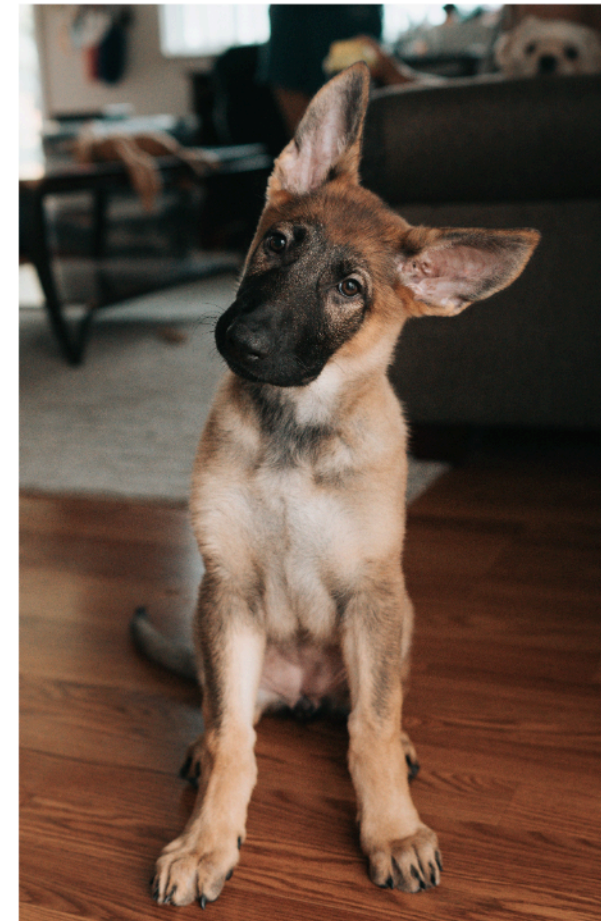
```
    report.md
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

- Git recognizes there are modified files not being tracked!

Nested repositories

- Don't create a Git repo inside another Git repo
 - Known as nested repos
- There will be two `.git` directories
- Which `.git` directory should be updated?



Let's practice!

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Staging and committing files

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The Git workflow

- Edit and save files on our computer
- Add the file(s) to the Git staging area
 - Tracks what has been modified
- Commit the files
 - Git takes a snapshot of the files at the point in time
 - Allows us to compare and revert files

Staging versus committing

Staging area



Making a commit



Adding to the staging area

- Adding a single file

```
git add README.md
```

- Adding all modified files

```
git add .
```

- `.` = all files in the current directory and sub-directories

Making a commit

```
git commit -m "Adding a README."
```

```
[main cb33c18] Adding a README.  
1 file changed, 1 insertion(+)  
create mode 100644 README.md
```

- `-m` Allows a log message without opening a text editor
- Log message is useful for reference
- Best practice = short and concise

Let's practice!

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